higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CARPENTRY AND ROOF WORK
NQF LEVEL 2

(12020122)

22 November 2019 (Y-Paper)
13:00–16:00

This question paper consists of 7 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Write neatly and legibly.

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QUESTION 1: BASIC TOOLS AND TERMINOLOGY

1.1 A spirit level is an important tool used in roof construction.

1.1.1 Name the parts of a spirit level. (2)

1.1.2 What is a spirit level used for? (1)

1.1.3 Name TWO other tools one could use instead of a spirit level. (2)

1.2 Give ONE advantage that a crowbar has over a claw hammer. (1)

1.3 Give ONE advantage and ONE disadvantage of using hand tools. (2)

1.4 State THREE precautions to take when working with power tools. (3)

1.5 List FIVE types of trusses used in roof construction. (5)

1.6 Explain the difference between *battens* and *purlins* with reference to their size and function. (2 + 2) (4)

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QUESTION 2: ROOF CONSTRUCTION

2.1 What is the main function of a roof? (1)

2.2 Name TWO types of loads acting on a roof. (2)

2.3 Give THREE advantages that a hipped roof has over a gable roof. (3)

2.4 Name and explain TWO criteria of roof design. (2 × 2) (4)

2.5 State FIVE principles/rules of roof construction. (5)

2.6 Briefly describe a wall plate and explain its purpose. (4)

2.7 When would it be necessary to support the trusses on the outer skin of brickwork? (1)

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2.8 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.8.1–2.8.5) in the ANSWER BOOK.

2.8.1 A maximum spacing of 1 000 mm, using 38 mm × 38 mm battens, is suitable for fibre-cement slates.

2.8.2 A truss spacing of 1 500 mm, using 50 mm × 76 mm purlins, must be used when covering a roof with fibre-cement corrugated sheets.

2.8.3 A 38 mm × 152 mm wall plate must be used when the roof span is between 10 m and 15 m.

2.8.4 Two wires of 2,4 mm diameter are used to tie down the roof trusses on a concrete tiled roof.

2.8.5 Hurricane clips are used to fix two timber members to each other at 90°.

(5 × 1) (5)

2.9 Give TWO requirements for roof anchors to be effective. (2)

2.10 List FOUR steps a carpenter would take when using a hoop iron to tie down roof trusses. (4)

2.11 Which type of nail is used for fixing lightweight steel purlins (eCo Purlins) to the rafters? (1)

2.12 Give THREE benefits of correctly transporting, handling and storing roof trusses. (3)

2.13 Give TWO advantages that bolted trusses have over nail-plated trusses. (2)

2.14 Give TWO reasons why washers are used in bolted truss connections. (2)

2.15 Name a device that can be attached to a crane to lift heavy trusses. (1)

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QUESTIONS 3: BUILD A HOWE ROOF TRUSS

3.1 Give the THREE stages of building a roof truss. (3)

3.2 Give ONE term for each of the following descriptions. Write only the term next to the question number (3.2.1–3.2.6) in the ANSWER BOOK.

3.2.1 Distance from the centre of the span to the outside of the wall plate

3.2.2 Angle between the top chord and the tie beam

3.2.3 Very top of the truss where the top chords meet

3.2.4 Part of the truss rafter that extends past the truss heel

3.2.5 Inside distance between the supporting walls

3.2.6 Top edge of a roof where the two slopes meet

(6 × 1) (6)

3.3 The king post of a Howe truss is fixed at 90° or square to the tie beam.

Explain how to use the 3:4:5 method to check for square when setting out a template for this truss. (8)

3.4 Name the TWO interior diagonal members common to a Howe truss. (1)

3.5 Name the type of metal fastener used to assemble a prefabricated truss. (1)

3.6 Which tool is used for marking the plumb cut on the rafters of a roof truss? (1)

[20]

QUESTION 4: GABLE ROOF CONSTRUCTION

4.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (4.1.1–4.1.10) in the ANSWER BOOK.

4.1.1 Safety signs on a building site that everyone can understand are essential because workers often speak ... language(s).

A one
B several
C different
D foreign

4.1.2 SANS 10243 stipulates that 12 mm bolts must have … washers.

A 25 mm × 25 mm × 2,4 mm round
B 32 mm × 32 mm × 6 mm square
C 25 mm × 25 mm × 2,4 mm square
D 36 mm × 36 mm × 6 mm square

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4.1.3 Nail plate connections can only be used in ... joints.

A bolted
B lap
C butt
D splayed

4.1.4 What is the maximum distance that a roof anchor can be positioned from a truss?

A 150 mm
B 100 mm
C 200 mm
D 50 mm

4.1.5 Timber used for roof construction must be … to prevent fungal and termite infestations.

A planed all around
B painted
C varnished
D treated

4.1.6 The NHBRC stipulates that in areas within 1 km from the coastline, nails must be treated by …

A painting.
B creosoting.
C preservation.
D galvanising.

4.1.7 A gable wall is usually ... a truss.

A much higher than
B the same shape as
C lower than
D thicker than

4.1.8 The NHBRC stipulates that the maximum distance of a truss adjacent to a gable wall should not be more than …

A 100 mm.
B 25 mm.
C 50 mm.
D 200 mm.

4.1.9 The law states that any company with more than ... employees must have a health and safety representative.

A 25
B 20
C 50
D 60

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4.1.10 The National Home Builders Registration Council stipulates that bolts used in a Howe truss must be ... bolts.

A Rawl
B coach
C machine
D hexagonal

(10 × 1) (10)

4.2 Rearrange the steps to erect trusses in the correct order by writing only the letters (A-J) in the correct order in the ANSWER BOOK.

A Use battens as temporary bracing to keep each truss in position.

B Lift the first and last trusses up onto the wall plates.

C Mark the position of all the other trusses.

D Fix a gut line at the apex of these trusses.

E Skew nail the first and last trusses to the wall plate.

F Check that all the trusses are evenly spaced.

G Fix stays to keep first and last trusses plumb and secure.

H Align the apex points of the trusses to the gut line and skew nail them to the wall plate.

I Mark the position of the first and last trusses on the wall plates.

J Lift all the other trusses up onto the wall plate.

(10 × 1) (10)
[20]

TOTAL: 100

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